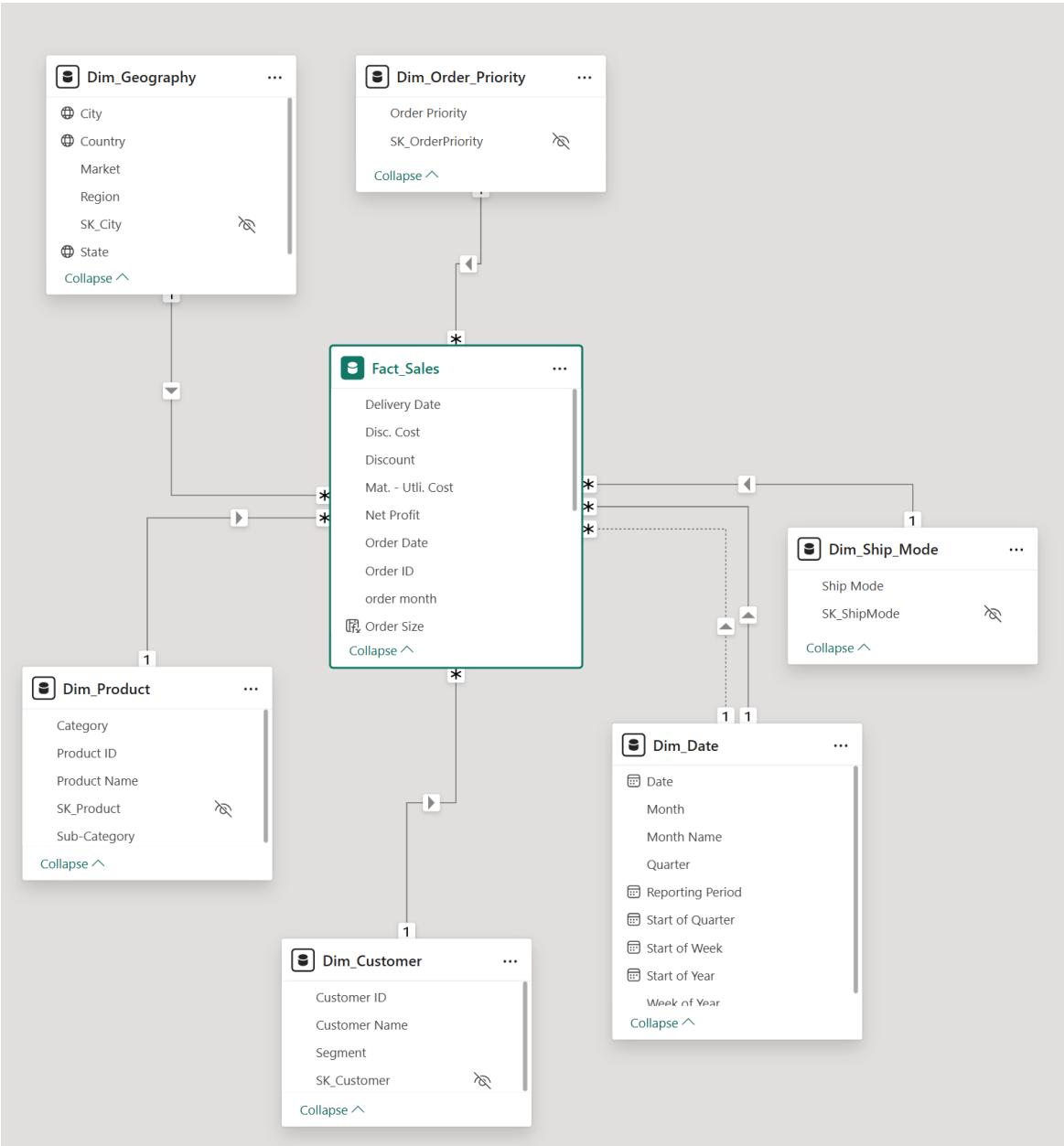


Star Schema Transformation Documentation

Introduction

This document outlines the transformation of a transactional dataset into a star schema model for improved performance and analytical efficiency in Power BI.

The star schema consists of a central fact table (**Fact_Sales**) linked to multiple dimension tables, enabling fast and scalable queries for business intelligence and reporting.



Star Schema Model Overview

The model consists of the following tables:

1. Fact_Sales (Fact Table)

The central table storing transactional sales data, containing:

- **Order ID** – Unique identifier for each order.
- **Order Date** – The date when the order was placed.
- **Delivery Date** – The date when the order was delivered.
- **Discount** – Discount applied to the order.
- **Disc. Cost** – Cost attributed to the applied discount.
- **Mat. - Util. Cost** – Material and utility cost of the order.
- **Net Profit** – The profit generated from the order.
- **Order Size** – Quantity of items in the order.
- **Order Month** – The month in which the order was placed.
- **Foreign Keys:** Links to dimension tables (**Customer ID, Product ID, City, Order Priority, Ship Mode, Date**).

2. Dim_Customer (Dimension Table)

Stores customer-related details:

- **Customer ID** – Unique identifier for the customer.
- **Customer Name** – Name of the customer.
- **Segment** – The business segment the customer belongs to (Consumer, Corporate, Home Office).
- **SK_Customer** – Surrogate key for customer.

3. Dim_Product (Dimension Table)

Stores product-related details:

- **Product ID** – Unique identifier for the product.
- **Product Name** – Name of the product.
- **Category** – Product category.
- **Sub-Category** – Sub-category of the product.
- **SK_Product** – Surrogate key for product.

4. Dim_Geography (Dimension Table)

Stores location-based details:

- **City** – The city where the order was placed.
- **State** – The state where the order was placed.
- **Country** – The country where the order was placed.
- **Market** – The market classification.
- **Region** – The region classification.
- **SK_City** – Surrogate key for geography.

5. Dim_Date (Dimension Table)

Stores date-related attributes for time-based analysis:

- **Date** – The date reference.
- **Month** – The numerical representation of the month.
- **Month Name** – The textual representation of the month.
- **Quarter** – The quarter of the year.
- **Reporting Period** – The relevant reporting period.
- **Start of Quarter** – The first day of the quarter.
- **Start of Year** – The first day of the year.
- **Week of Year** – The week number within the year.

6. Dim_Ship_Mode (Dimension Table)

Stores details about shipping methods:

- **Ship Mode** – The type of shipping method used (e.g., Standard Class, First Class, Same Day).
- **SK_ShipMode** – Surrogate key for ship mode.

7. Dim_Order_Priority (Dimension Table)

- **Order Priority** – Priority level of the order (e.g., High, Medium, Low, Critical).
- **SK_OrderPriority** – Surrogate key for order priority.